

Physics question bank for the third formative test

Unit 11: Explain evidences of modern physics, X- rays and Laser		الوحدات التدريبية
TPK01 – TPK02		أكواد المعارف
Unit 11: Explain evidences of modern physics, X- rays and Laser		المعارف
Radiation of Black Body and The Phenomenon Of The Thermal Ion Emission And Photoelectric	TPK01	
Compton Scattering and X- Rays & LASER	TPK02	

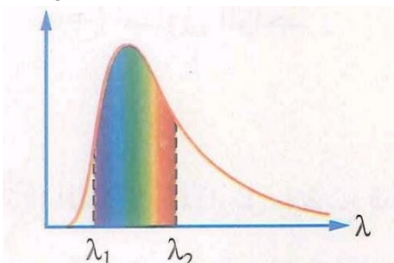
Unit 11	Lesson one Lesson two Lesson three	Lesson four Lesson five Lesson six
Knowledge code	TPK01	TPK02

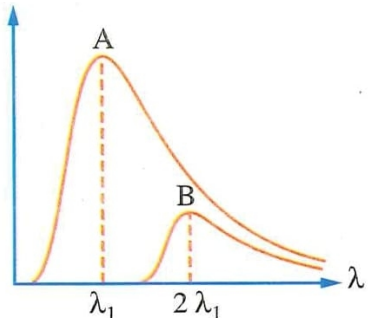
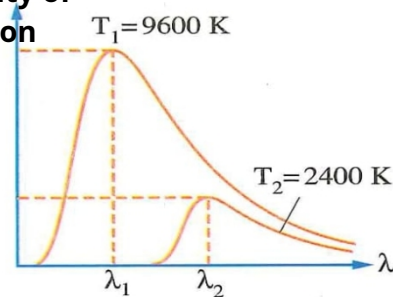
المعرفة	الكود
Radiation of Black Body and The Phenomenon	TPK01

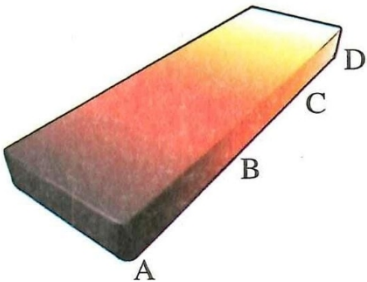
1- Choose the correct answer:

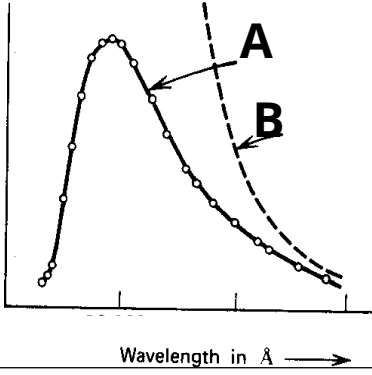
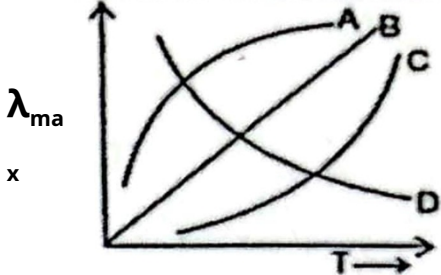
#	Question	Choices	Answer
1	Which of the following electromagnetic waves have the shortest wavelength	a. IR b. X-Rays c. UV d. Visible Light	
2	The electromagnetic radiation that have the highest frequency and the shortest wavelength.....	a. Radio waves. b. Visible waves. c. Gama rays .	
3	Due to the classical expectation the intensity of radiation increases by	a. Decreasing wavelength of radiation b. Decreases frequency of radiation c. Increases wavelength of radiation	
4	the wavelength of a photon is 2.2×10^{-7} m, its energy is given by	a. 1×10^{35} joule b. 9×10^{-19} joule c. 3.3×10^{-27} joule d. 4.4×10^{-32} joule	
5	Difference between modern physics and quantum physics	a. Modern physics talks about macroscopical model	

		b. Quantum physics talks about macroscopical model c. Both are the same	
6	Difference between modern physics and classical physics	a. Modern physics talks about macroscopical model b. Classical physics talks about macroscopical model c. Both are the same	
7	The emitted energy of the sun radiation is ----- visible light.	a. 40% b. 50% c. 20% d. 80%	
8	The emitted energy of the sun radiation is ----- heat radiation.	a. 40% b. 50% c. 20% d. 80%	
9	The emitted energy of the lamp radiation is ----- visible light.	a. 40% b. 50% c. 20% d. 80%	
10	The emitted energy of the lamp radiation is ----- heat radiation.	a. 40% b. 50% c. 20% d. 80%	
11	The electromagnetic radiation that have the lowest frequency and the tallest wavelength.....	a. Radio waves. b. Visible waves. c. Gama rays .	
12	It explains all phenomena involved in electronics which is the basis for all modem electronic and communication systems.	a. Classical physics. b. Quantum physics. c. All the above .	
13	The speed of electromagnetic waves equals	a. $3 \times 10^8 \frac{m}{s}$ b. $3 \times 10^{-8} \frac{m}{s}$	

		c. $4 \times 10^8 \frac{m}{s}$																
1 4	The speed of electromagnetic waves is calculated by the relation	a. $C = \cancel{\lambda} v$ b. $C = \cancel{\lambda} \cancel{v}$ c. $C = \cancel{\lambda} - v$																
1 5	if the body is ideal black body , so its dominant color for the radiation that radiate from it is changes according to	a. the mass of the body. b. The density of the body. c. The temperature of the body. d. The type of the material that make the body.																
1 6	In the plank's curve for the black body radiation, if the temperature decreased	<table><tr><td></td><td>The peak of the curve</td><td>The space below the curve</td></tr><tr><td>A</td><td>Displaced to the high frequency region</td><td>Increase</td></tr><tr><td>B</td><td>Displaced to the high frequency region</td><td>Decrease</td></tr><tr><td>C</td><td>Displaced to the low frequency region</td><td>Increase</td></tr><tr><td>D</td><td>Displaced to the low frequency region</td><td>Decrease</td></tr></table>		The peak of the curve	The space below the curve	A	Displaced to the high frequency region	Increase	B	Displaced to the high frequency region	Decrease	C	Displaced to the low frequency region	Increase	D	Displaced to the low frequency region	Decrease	
	The peak of the curve	The space below the curve																
A	Displaced to the high frequency region	Increase																
B	Displaced to the high frequency region	Decrease																
C	Displaced to the low frequency region	Increase																
D	Displaced to the low frequency region	Decrease																
1 7	In the Planck's curve for the black body radiation, the intensity of radiation at the very high frequencies is	a. Doesn't change. b. Moderate. c. Very high. d. Close to zero.																
1 8	At the opposite graph, if λ_1 is the shortest wavelength of the visible rays, while λ_2 is the longest wavelength for visible rays, so the graph might represent the radiation that Intensity of radiation 	a. A glowing star. b. Earth. c. Tungsten lamp. d. Human body.																

	radiate from		
1 9	The opposite graph represents the relation between the intensity of radiation and the wavelength for two hot bodies A,B, so the ratio Intensity of radiation  between their temperatures ($\frac{T_A}{T_B}$) is	a. $\frac{4}{1}$ b. $\frac{2}{1}$ c. $\frac{1}{2}$ d. $\frac{1}{4}$	
2 0	The opposite graph shows a hot black body at T_1, T_2 so the ratio between ($\frac{\lambda_1}{\lambda_2}$) is Intensity of radiation 	a. $\frac{1}{16}$ b. $\frac{1}{8}$ c. $\frac{1}{4}$ d. $\frac{1}{2}$	

2 1	The opposite figure shows a hot iron piece, at which point that has the lowest temperature..... 	a. A b. B c. C d. D	
2 2	A black body radiation with 3000 K, and its maximum wavelength λ, if the body is cooled to be T, its maximum wavelength become 10λ, so T equals	a. 300 K. b. 2700 K. c. 270 K. d. 1800 K.	
2 3	In Planck's curve, the wavelength associating the maximum intensity of radiation emitted from the sun lies in region of	a. UV b. V.R c. I.R d. X-rays	
2 4	The frequency of photons which emitted from a glowing body increases, then its number	a. Increases b. Decreases c. Remains constant d. Equals zero	
2 5	Which of the following curves represents the classical explanation for blackbody radiation, and which one is the modern explanation	a. A: modern B: classical. b. A: classical B: classical. c. A: modern B: modern. d. A: classical B: modern .	

			
2 6	The distribution of the radiation intensity with wavelength is	a. Wein's distribution. b. Planck's distribution. c. Classical physics distribution. d. Modern physics distribution.	
2 7	The wavelength at maximum radiation intensity is inversely proportional to the temperature is	a. Blackbody radiation. b. Wein's law. c. Planck's law.	
2 8	For a black body radiation if $\lambda_{\max}$ represents the maximum wavelength corresponding to maximum radiation at temperature T, the curve between $\lambda_{\max}$ and T is 	a. A b. B c. C d. D	

2- Mention the factor affecting on

No.	Question	Answer
1	Energy of the photon	
2	The color of the hot objects	
3	The intensity of the black body radiation	
4	The wavelength at maximum radiation intensity (λ_{max})	
5	The speed of electromagnetic waves in air	
6	The speed of electromagnetic waves in media	

3- True or False

No.	Question	Answer	Correction if F
1	when a piece of iron is heated to become self-luminous, the apparent color of the emitted radiation shifts initially from red to orange to yellow to blue and finally to white as the temperature increases		
2	The sun is yellow because its surface temperature is about 6000 K. Some stars glow blue-white because their surface temperature can be as high as 20000K		
3	Wave particle duality states that a wave can only act as a wave and not a particle.		
4	Visible light is the part that can be seen with the human eye.		
5	Visible light is only a very large part of the electromagnetic spectrum.		
6	Black body reflects no radiation and appears perfectly black.		
7	In Wien's law, the smaller temperature, the smaller the wave length of the peak		
8	All objects radiate energy continuously in the form of electromagnetic waves		

4- Mention the scientific term

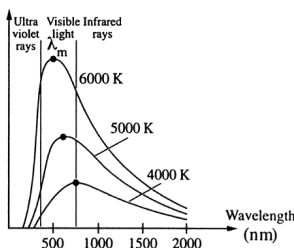
No.	Question	Answer
1	It is the branch of physics that explains everything in our daily life and our common experiences	
2	It is the branch of physics that deals with scientific phenomena in the universe which might not be directly observed in our daily life, and classical physics cannot explain	
3	They are vibrating electric and magnetic fields that moves through space or any other medium	
4	The wavelength at maximum radiation intensity (λ_{max}) is inversely proportional to the temperature	
5	The wavelength (λ_{max}) at which the peak of the curve occurs is inversely proportional to temperature	
6	It is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again	
7	The distribution of the radiation intensity with wavelength	
9	A body that absorbs all the radiation falling on it with different wavelengths the emits them again in an ideal form .	
10	The transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet radiation.	
11	the ratio between the photon energy to its frequency.	
12	A phenomenon used for criminology and night vision	

5- complete the following sentences:

No.	Question	Answer
1	is the transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet radiation.	
2	 is the physics that explains everything in our daily life and our common experiences.	
3	is resulting from vibrating electric and magnetic fields that moves through space or any other medium.	
4	 is the part that can be seen with the human eye.	
5	There are two types of body according to its radiation,,and	
6	is that deals with scientific phenomena in the universe which might not be directly observed in our daily life	
7	The wave particle duality means that the wave can behave as a,and	
8	All objects radiate energy continuously in the form of produced by thermal vibrations of the molecules.	
9	 deal with atomic and subatomic systems.	
10	All phenomena involved in electronics which is the basis for all modern electronic and communication systems is explained by	

11	its wavelength λ ranges from 400nm to 700nm.	
12	Above 400nm is called, while below 700nm is called	
13	According to $E=h\nu$, E is Proportional with ν .	
14	The atom release gained energy in the form of when it return into its original orbit .	
15	Radiation is composed of small particles called	
16	40% of The sun radiation, and 20% of The lamp radiation is	
17	50% of The sun radiation, and 80% of The lamp radiation is	
18	The maximum intensity of radiation increases by increasing According to Planck's Distribution.	
19	The wave length at maximum radiation intensity (λ_{max}) is..... proportional to the temperature.	
20	 is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again	
21	is a Prefect absorber and prefect emitter.	
22	Classical explanation for black body radiation states that The intensity of radiation increases with the increasing of	
23	Major disagreement exists between classical theory and modern physics at	

6- Solve the following:

No	Question	Answer
1	If the temperature of surface of sun is 6000 K and wavelength (λ_{max}) at maximum radiation intensity is 0.499 nm. Calculate the temperature of filament lamp that gives photons at wavelength (λ_{max}) = 1.2 nm	
2	if the wavelength having a maximum intensity in the solar radiation is 0.5 μm Calculate the wavelength having a maximum intensity in the radiation emitted by Earth, given that the surface temperature of the sun and Earth are 6000 K and 300 K respectively.	
3	calculate the wavelength of photon with energy = 4.968×10^{-19} Joule	
4	A body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å, calculate the temperature of the sun	
5	What is the energy of a photon with frequency $3.0 \times 10^{14} \text{ Hz}$	
6	The wavelength of yellow light is approximately 5.9×10^{-7} meters. What is its frequency	
7	(a) What are the names of these curves? (b) Does the classical physics succeeded to explain these curves? And why? (c) What is meant by (λ_m)? (d) What will happen to (λ_m) when the temperature decreases? 	(a)

7- Compare between the following:

1-

In terms of	Emitted rays from the sun	Emitted rays from the Earth
Region of maximum radiation		

2-

In terms of	Emitted rays from the sun	Emitted rays from the electric lamp
Temperature		
The wavelength at the peak		
the emitted energy		

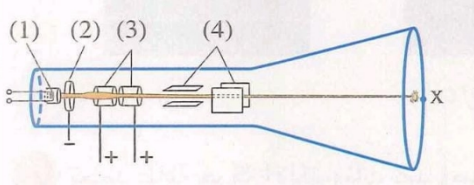
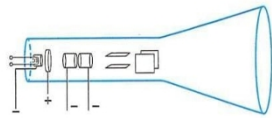
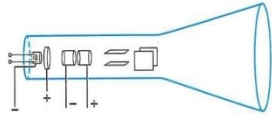
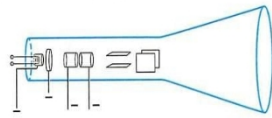
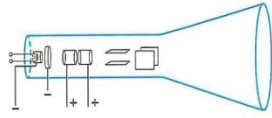
3-

In terms of	Classical physics	Modern physics
Definition		

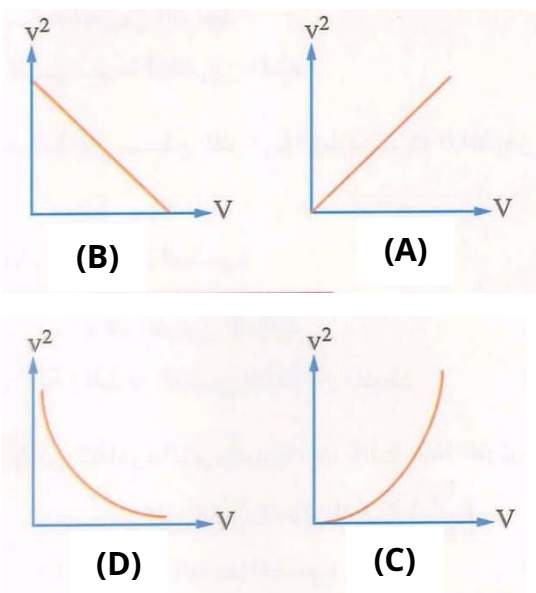
المعرفة	الكود
Thermionic emission	TPK01

1- Choose the correct answer:

#	Question	Choices	Answer
1	The cathode rays consist of	a. Electrons b. Photons c. Electromagnetic Wave d. Gamma	
2	Cathode rays have the energy of	a. $\frac{1}{2} m_e V^2$ b. $m_e V$ c. $2 m_e V$	
3	The process in which electrons are emitted from a metallic surface when heat given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	
4	CRT stands for	a. Common Ray Tube b. Common Running Tube c. Cathode Ran Tube d. Cathode Ray Tube	
5	In the C.R.T when the grid voltage changes from -20 V to -50 V.....	a. Intensity of illumination increases on fluorescent screen. b. Intensity of illumination decreases on fluorescent screen. c. The deflection of electron beam increases. d. The deflection of electron beam decreases.	
6	From the properties of beam of C.R.T.....	a. Electromagnetic waves. b. With constant speed.	

		c. Charged particles. d. Uncharged particles.	
7	The opposite figure represents C.R.T, which of the following parts represents..... A. The source of electrons..... B. The part that responsible for changing the position of collision of electron beam with fluorescent screen..... C. The illumination intensity is controlled by At point X.	(A): a. Part (1). b. Part (2). c. Part (3). d. Part (4). (B): a. Part (1). b. Part (2). c. Part (3). d. Part (4). (C): a. Part (1). b. Part (2). c. Part (3). d. Part (4).	
			
8	Which of the following figures represents the correct voltage that is used in C.R.T.....	 (B)  (A)  (D)  (C)	
9	In the C.R.T when the filament is damaged	a. Intensity of illumination increases on fluorescent screen. b. Intensity of illumination decreases	

		on fluorescent screen. c. The fluorescent screen doesn't illuminate. d. The deflection of electron beam decreases.	
1 0	In the C.R.T when the grid is not connected to any electric signal	a. It is hard to control the path of electric beam into the screen. b. The fluorescent screen doesn't illu c.  (B) d.  (A)	
1 1	Which of the following figures represents the final image that formed in the screen of C.R.T when the deflection system doesn't work.....	 (D)  (C)	
1 2	The kinetic energy for the electrons is depend on when it reaches the Anode in C.R.T.	a. The surface area of cathode. b. Work done for Anode. c. Potential difference between two plates of deflection system. d. Potential difference between cathode and anode.	
1 3	In C.R.T the velocity of electron is v , when the electron is accelerated by potential difference with V , if the potential difference is increased to $2V$, son the velocity will become.....	a. $2 v$ b. $\sqrt{4 v}$ c. $4 v$ d. $1\sqrt{2} v$	

1 4	Which of the following graphs represents the relationship between v^2 for the electron and V in C.R.T.....	 (B) (A) (D) (C)	
1 5	If the potential difference between anode and cathode in C.R.T is 1000 V : so,... A. K.E for the electron is B. The maximum velocity of liberated electron from cathode when it reaches anode equals.....	(A): a. $1.6 \times 10^{-15} J$ b. $1.6 \times 10^{-16} J$ c. $1.6 \times 10^{-17} J$ d. $32 \times 10^{-18} J$ (B) : a. $1.88 \times 10^6 m/s$ b. $1.88 \times 10^7 m/s$ c. $3.52 \times 10^5 m/s$ d. $3.52 \times 10^4 m/s$	
1 6	The source of thermal energy in the C.R.T is.....	a. Anode. b. Cathode. c. Filament. d. Fluorescent screen.	
1 7	The reason for the kinetic energy of the electron is.....	a. Velocity. b. Density c. Mass of the electron. d. Potential difference between cathode and anode.	
1 8	It is used to accelerate the electron beam towards the screen.....	a. Cathode. b. Anode. c. Grid. d. Filament.	

19	To increase to velocity of electron in C.R.T for four times.....	a. Increase K.E for four times. b. Increase the mass of electron to double. c. Increase the potential difference to double. d. Increase the potential difference for four times.	
20	To decrease the intensity of electron beam	a. Increase the voltage of grid. b. Decrease the voltage of anode. c. Increase the voltage of anode. d. Decrease the voltage of grid.	
21	Which part of C.R.T is responsible for the deflection of electron beam.....	a. Vertical plates. b. Horizontal plates. c. Anode. d. A,B.	

2- Mention the application of

No	Question	Answer
1	The electron gun in CRT	
2	CRT	
3	Deflection system in CRT	
4	fluorescent screen in CRT	
5	Filament in CRT	
6	Cathode in CRT	
7	Grid in CRT	
8	Anode in CRT	

3- Mention the factor affecting on

No.	Question	Answer
1	KE of the freed electron in CRT	
2	the velocity in freed electron in CRT	

4- Solve the following:

No.	Question	Answer
1	An electron moving at a constant speed 4.4×10^5 m/sec, what is its kinetic energy?	
2	Calculate the speed of an electron that has kinetic energy of 120 eV.	
3	calculate the potential difference which accelerated an electron from rest with speed equals 9.796×10^6 m/s	

5- complete the following sentences:

No.	Question	Answer
1	There are many ways to eject (escape) electrons out of a material,and	
2	Emission of electrons from a heated metal's surface (by heating) is called	
3	 is application on the thermionic emission.	
4	C.R.T is used to make and Monitors.	
5	 is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it.	
6	Operation stages for C.R.T are	

	,and	
7	 is used to control the number of electrons of the electron beam .	
8	 is used to accelerate the electron beam towards the screen.	
9	The stage at which the beam at high speed.....	
10	emits green light when the electron strike it.	

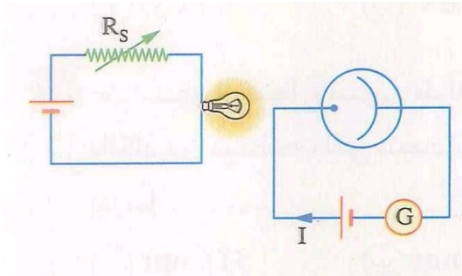
6- Mention the scientific term

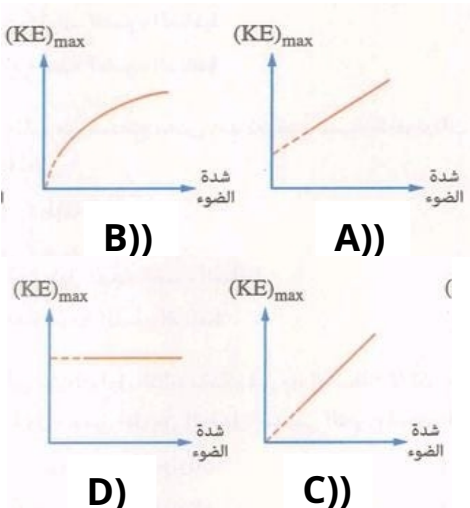
No.	Question	Answer
1	emission of electrons from a heated metal's surface (by heating)	
2	emission of electrons from a metallic surface when the light energy falling on it	
3	It is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it	

المعرفة	الكود
Photoelectric effect	TPK01

1-Choose the correct answer:

#	Question	Choices	Answer
1	The process in which electrons	a. Photoelectric effect	

	are emitted from a metallic surface when a light given to it is called	b. Thermionic effect c. Compton effect d. Black body radiation	
2	the emission of electrons from metallic surface (photoelectric effect) depends on	a. The intensity of the incident light b. The frequency of incident light c. The exposure time	
3	The process in which electrons are emitted from a metallic surface when a light given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	
4	The emission of electrons from cathode in photoelectric cell is depend on	a. The type of material of Anode and the light intensity. b. The type of material of Anode and the light frequency. c. The type of material of Cathode and the light intensity. d. The type of material of Cathode and the light frequency.	
5	At the opposite figure shows light produced from an electric lamp falls on a photoelectric cell cases flow of photocurrent, if the light intensity increases, the photocurrent 	a. Increases. b. Decreases. c. Vanishes. d. Doesn't change.	

6	Which of the following graphs represents the relationship between $K.E_{\text{Max}}$ for emitted photoelectrons and light intensity	 The figure contains four graphs labeled A, B, C, and D. Each graph has $(KE)_{\text{max}}$ on the vertical axis and light intensity (شدة الضوء) on the horizontal axis. Graph A shows a straight line starting from the origin. Graph B shows a curve starting from the origin and increasing with a decreasing slope. Graph C shows a straight line starting from a positive value on the vertical axis. Graph D shows a horizontal line at a positive value on the vertical axis.	
7	An electromagnetic radiation is incident on a photoelectric cell with ν , photoelectrons emitted with $K.E_{\text{Max}}$, if the frequency becomes 2ν , which of the following doesn't change	a. The photon energy. b. Speed of photon c. $K.E_{\text{Max}}$ for emitted photoelectron. d. Maximum velocity of emitted photoelectron.	
8	A monochromatic light incident on photoelectric cell with frequency greater than the critical frequency for cathode, then another light is used with the same wavelength of the first one but its intensity is double of the first, which one is doubled	a. The energy of each photon e. $K.E_{\text{Max}}$ for emitted photoelectron. b. Work function for the metal. c. The intensity of photocurrent.	
9	If the frequency of incident light increase to double, so the work function	a. Increase to double b. Decrease to half c. Remains constant d. Increase for four times.	
10	When the incident light has wavelength 623 nm on a metal, photoelectrons are emitted with 4.6×10^5 m/s, so : A- The work function of the	(A) a. $2.23 \times 10^{-19} \text{ J}$. b. $3.19 \times 10^{-19} \text{ J}$. c. $4.15 \times 10^{-19} \text{ J}$. d. $4.6 \times 10^{-19} \text{ J}$. (B)	

	metal B- The critical frequency of the metal	a. 6.94×10^{14} b. $6.26 \times 10^{14} \text{ HZ.}$ c. $4.81 \times 10^{14} \text{ HZ.}$ d. $3.34 \times 10^{14} \text{ HZ.}$	
11	If the critical frequency of a metal equals the frequency of the blue light , which of the following rays will be able to emit the electrons from the metal surface.....	a. Infrared rays. b. Radio waves. c. Red light. d. Ultraviolet rays.	
12	If a light with $3100 \text{ \AA}$ incident on the surface of cathode in photoelectric cell, so the electrons are emitted with kinetic energy equals 2.5 eV , so the work function for the cathode is	a. 3.1 eV. b. 2.4 eV. c. 1.5 eV. d. 0.9 eV.	
13	If the work function for a metal equals $3.3125 \times 10^{-19} \text{ J}$, the critical frequency is	a. $4.5 \times 10^{14} \text{ HZ.}$ b. $4.8 \times 10^{14} \text{ HZ.}$ c. $5 \times 10^{14} \text{ HZ.}$ d. $5.5 \times 10^{14} \text{ HZ.}$	

2- Mention the factor affecting on

No.	Question	Answer
1	KE of the freed electron by photoelectric effect	
2	the velocity of the freed electron by photoelectric effect	

3- True or False

No.	Question	Answer	Correction if F
1	Changing the color of the light means we are changing the frequency.		
2	In Photoelectric effect, electrical energy converted to light energy		
3	Photoelectrons are electrons that are released or ejected from a substance by photoelectric effect		
4	Kinetic energy is independent of the intensity of light.		
5	Increasing the brightness of the light, increases the speed of the electrons.		
6	Increasing the intensity of the light, increases the number of electrons.		
7	increasing the frequency, increases the kinetic energy.		
8	Photoelectric effect is always instantaneous		
9	The photon energy doesn't have to be more than the work function for electrons to be ejected.		

4-Solve the following:

No.	Question	Answer
1	a monochromatic radiation of wavelength 3×10^{-7} m falls on a metal surface, so electrons liberated with maximum kinetic energy 3.26×10^{-19} J a) the energy of the photon that falls on the metal surface b) the work function of the metal	
2	photon with wavelength 4×10^{-7} m falls on a surface of metal with work function 2.3×10^{-19} J, find the kinetic energy of emitted electron from a surface of the metal	

3	A metal has a work function 8×10^{-19} J. Calculate the maximum kinetic energy of electrons emitted from the metal when light of frequency 2×10^{15} Hz is shone on the surface.	
4	When monochlight of wave length 663 nm is incident on a metal plate whose work function 3.52×10^{-19} eV will electrons be emitted or not?	
5	Draw the relationship between light intensity and photocurrent at the case of $E < E_w$ and $E \geq E_w$.	(a)

5- complete the following sentences:

No.	Question	Answer
1	There are many ways to eject (escape) electrons out of a material are and	
2	The emission of electrons from a metallic surface when the light energy falling on is	
3	The electrons that released or ejected from a substance by photoelectric effect is called	
4	An application on photoelectric effect is	
5	photoelectric cell aims to convert To	
6	the source of free electrons in photoelectric cell is	
7	the collector in photoelectric cell is called	

		
8	the minimum energy needed to free (release) the electrons from the surface of the metal is	
z	To free (emit) an electron with kinetic energy from the metal, the energy of the incident photon must be..... .	
10	The kinetic energy of the emitted electrons is independent of	
12	At the case of the relationship between light intensity and photocurrent is independent of each other.	
13	At the case of the relationship between light intensity and photocurrent is directly proportional to each other.	

5- Mention the scientific term

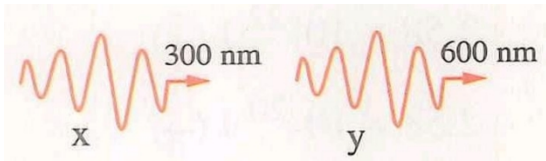
No.	Question	Answer
1	An electron released or ejected from a substance by photoelectric effect	
2	It is the minimum energy needed to free (release) the electrons from the surface of the metal	

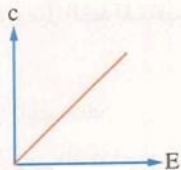
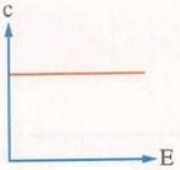
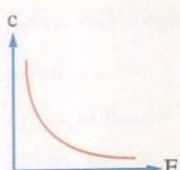
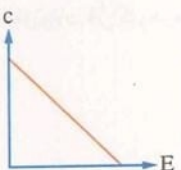
المعرفة	الكود
Compton Scattering	TPK02

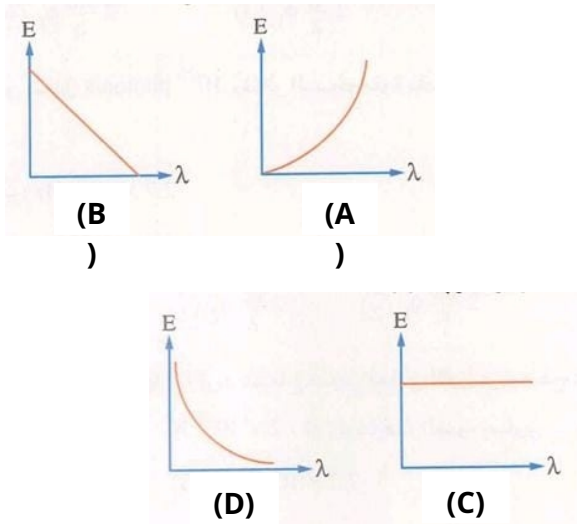
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1-Choose the correct answer:

#	Question	Choices	Answer
1	The ratio between the energy of the photon to the velocity of light in air is the of the photon	a. Mass b. Frequency c. Kinetic energy d. Momentum	
2	the ratio of the photon's momentum to its mass equals	a. The velocity of light b. Planck's constant c. The photon energy d. photons are massless particles	
3	in Compton effect, ratio between the photon energy after collision to its energy before collision is	a. Greater than one b. Equal one c. Less than one	
4	In Compton effect, the ratio between the velocity of photon after collision to the velocity of photon before collision	a. Greater than one b. Equal one c. Less than one	
5	In Compton effect, the ratio between the frequency of photon after collision to the frequency of photon before collision	a. Greater than one b. Equal one c. Less than one	
6	In Compton effect, the ratio between the wavelength of photon after collision to the wavelength of photon before collision	a. Greater than one b. Equal one c. Less than one	
7	in Compton effect, the speed of gamma rays photons after collides with a free electron is	a. Increased b. Decreased c. Equal zero d. Does not change	
8	in Compton effect, gamma photons is scattered and its increases	a. Energy b. Speed c. Wavelength	

		d. Momentum	
9	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon frequency decreased and changed its direction; also the electrons velocity increased and changed its direction	a. Characteristic radiation b. Continuous radiation c. Compton effect d. Thermionic emission	
10	The Compton effect proves	a. The particle nature of photon. b. The wave nature of photon. c. The wave nature of electron. d. The particle nature of electron .	
	In Compton effect, when a photon with high frequency collides with free electron , which of the following will increase for photon after collision	a. Energy . b. Frequency. c. Wavelength. d. Linear momentum.	
	In Compton effect after collision , there is no decrease of	a. The wavelength of electron. b. Energy of photon. c. Photon frequency. d. Photon speed.	
	From the properties of photon	a. Its speed equals the speed of light . b. Deflects by the electric field. c. It may be accelerated. d. All the above.	
	At the opposite figure, the ratio between the speed of photon x and y is..... 	a. $\frac{4}{1}$ b. $\frac{2}{3}$ c. $\frac{1}{1}$ d. $\frac{1}{3}$	
	The linear momentum for a photon is	a. $\frac{h}{\lambda}$	

		b. $\frac{hv}{\lambda}$ c. $\frac{hc}{\lambda}$ d. $\frac{hv}{c^2}$	
	Photon x has P_L while photon y has $2 P_L$, so	a. The wavelength of x is double wavelength of y. b. The wavelength of x is half wavelength of y. c. The speed of x is double speed of y. d. The speed of x is half speed of y.	
	Two photon the ratio between their frequencies is 1:2, so the ratio between their energies is	a. 1:1 b. 1:2 c. 2:1 d. 4:1	
	A monochromatic light its number of photons is n and its wavelength is λ , so the total energy is	a. $\frac{hc}{n\lambda}$ b. $\frac{n\lambda}{hc}$	
	Which of the following graphs represents the relation between the speed of photon and its energy	 (B))  (A))  (D)  (C)	

	Which of the following graphs represents the relation between the energy of photon and its wavelength.....	 (A) (B) (C) (D)	
	Photons are electrically neutral particle and they are not deflected by the either electric or magnetic field.....	a. True b. False c. Non of the above	
	1. According to Compton effect when a photon collides with a free electron. Then,	a. the photon loses its energy. b. the electron gains a kinetic energy equals that of the photon energy. c. the frequency of the scattered photon is less than the frequency of the incident photon. d. the electron and the photon move together on the same line.	
	In Compton experiment , the sum of the photon energy and the electron energy before collision is the sum of their energies after collision.	a. greater than b. less than c. equal to d. no correct answer	
	In Compton effect, the ratio between the velocity of photon after collision to the velocity of photon before collision.....	a. Greater than one b. Equal one c. Less than one	

2- True or False

No.	Question	Answer	Correction if F
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1	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, The photon's frequency decreases		
2	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, The photon's wavelength decreases		
3	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed decreases.		
4	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed increases.		
5	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed remains the same.		
6	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the electron's speed increases.		
7	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the electron's speed decreases.		
8	Classical theories can explain the change in wavelength.		
9	The conservation of energy states that energy of the photon before and after collision is the same.		
10	The conservation of energy states that energy of the electron before and after collision is the same.		
11	The conservation of energy states that energy of the sum of energy of photon and electron before and after collision is the same.		

3-Solve the following:

No .	Question	Answer
1	A photon of X rays of ($\lambda = 3 \text{ \AA}$) collides a stationary electron, the electron is scattered with energy $1.1 \times 10^{-16} \text{ J}$, so the wavelength of the	

	scattered photon is $\text{\AA}$	
2	calculate the photon frequency, mass and linear momentum if $\lambda = 380 \text{ nm}$	
3	the energy of photon $3.3 \times 10^{-19} \text{ joule}$. Calculate its mass, frequency and wavelength.	
	From studying Compton effect, mention what happens after collision for each value of the following, then mention the reason : (a) The photon energy. (b) The photon velocity.	(1 st session 10)

4- Mention the scientific term

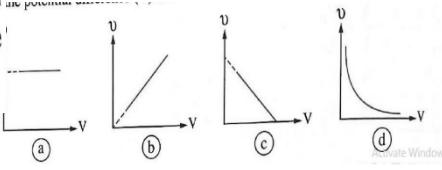
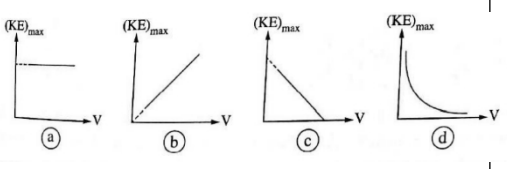
No.	Question	Answer
1	The total linear momentum is the same before and after collision	
2	The total energy is the same before and after collision	

المعرفة	الكود
X-Ray	TPK02

1-Choose the correct answer:

#	Question	Choices	Answer
1	The wavelength of x-rays range is ... in $\text{\AA}$	a. 10^{-13} to 10^{-8} b. 10^{-13} to 10^2 c. 10^{-3} to 10^2 d. 10^{-3} to 10^{-8}	
2	The wavelength of x-rays range is ... in meters	a. 10^{-13} to 10^{-8} b. 10^{-13} to 10^2	

		c. 10^{-3} to 10^2 d. 10^{-3} to 10^{-8}	
3	This radiation is called soft X-rays	a. Characteristic radiation b. Continuous radiation c. Both a and b	
4	This radiation is sometimes called bremsstrahlung	a. Characteristic radiation b. Continuous radiation c. Both a and b	
5	Bremsstrahlung is a ... word	a. Latin b. Swedin c. German	
6	Bremsstrahlung means	a. braking radiation b. breaking radiation c. barking radiation	
7	The x-ray is considered	a. Visible electromagnetic wave. b. Invisible electromagnetic wave. c. Mechanical wave. d. UV ray.	
8	The tube that is used to produce x-ray	a. C.R.T b. Photocell c. Coolidge tube d. Resonant cavity.	
9	The filament in the Coolidge tube is responsible for.....	a. Source of heat energy. b. Source of free electrons. c. A,B. d. None of these.	
10	The cooling fins are responsible for.....	a. Reduce light energy. b. Reduce x-rays. c. Reduce heat energy. d. Reduce electric energy.	
11	The spectrum of the x-rays is	a. Continuous spectrum. b. Line spectrum. c. Both .	
12	The line spectrum resulted from	a. Characteristic radiation. b. Thermal radiation.	

		c. Soft radiation. d. Braking radiation.	
1 3	The continuous spectrum resulted from....	a. Characteristic radiation. b. Thermal radiation. c. Black body radiation. d. Braking radiation.	
1 4	The Characteristic radiation is depend on	a. the kind of target element. b. the voltage difference. c. None of these	
1 5	The continuous radiation is depend on	a. the kind of target element. b. the voltage difference. c. None of these.	
1 6	Which of the following graphs represents the relation between the maximum frequency (ν) for the photons of the continuous spectrum of the X-rays which are generated from the Coolidge tube and the potential difference (V) between the anode and the cathode?		
1 7	In the Coolidge tube, if the potential difference between the cathode and the anode doubled, then the shortest wavelength for the braking spectrum of the X-rays	(a) does not change (b) decreases to quarter its initial value (c) decreases to half its initial value (d) is doubled	
1 8	Which of the following graphs represents the relation between the maximum kinetic energy ($(KE)_{max}$) acquired by one of the electrons which is emitted from the cathode in the Coolidge tube and the potential difference (V) between the cathode and the anode?		

2- True or False

No.	Question	Answer	Correction if F
1	The continuous spectrum produces all wavelengths within a certain range regardless of the target material		
2	Continuous radiation is generated when an electron collides with an electron close to the nucleus of the target material atom.		

3	The wavelength of the characteristic radiation (at the peak) does not depend on the voltage difference		
4	The wavelength of the characteristic radiation (at the peak) does not depend on the kind of target element		
5	The wavelength of the characteristic radiation (at the peak) depends on the voltage difference		
6	The higher the atomic number of the target element, the shorter the wavelength of the characteristic radiation		
7	At low voltage differences, no characteristic radiation is produced.		
8	The characteristic radiation is generated when the velocity of the colliding electrons is reduced upon passing through a retarding electric field due to the electrons of the target material.		

3-Mention the application of

No.	Question	Answer
1	Coolidge tube	
2	Cooling fin in Coolidge tube	
3	vacuum tube in Coolidge tube	
4	An external high DC voltage source in an external high DC voltage source	
5	Filament in Coolidge tube	

4- Mention the factor affecting on

No.	Question	Answer
1	the minimum wavelength of x-rays of continuous radiation	
2	characteristic x-rays radiation	

5-Solve the following:

No.	Question	Answer
1	X-rays emits by wavelength 632 nm, what is the frequency of x-rays in Hz	
2	calculate the maximum frequency of the generated X rays from Coolidge tube when the voltage difference equal to 10000 volt	
3	X-rays are generated in the x-rays tube with operating voltage of 20kv. Calculate: a) the maximum energy of the emitted photons b) the maximum frequency of the emitted photons	
4	State the applications of x-ray	a.
5	State the properties of X-Ray	a.

المعرفة	الكود
LASER	TPK02

1-Choose the correct answer:

#	Question	Choices	Answer
1	laser beam production is based on the principle of	a. Induced absorption b. spontaneous emission c. ionization d. simulated emission	
2	from the characteristic of laser beam	a. monochromatic b. non coherence c. varying in phase d. all of the above	
3	laser light is monochromatic because	a. It has approximately one	

		wavelength b. Its photons are of the same phase c. It is parallel rays d. all of the above	
4	laser beam has high intensity, which means	a. It doesn't obey inverse square law b. It has only one wavelength c. Its photons have the same direction d. all of the above	
5	which of the following is not true for laser	a. Extremely intense light b. Perfectly monochromatic c. Coherent d. Following inverse square law	
6	laser beam doesn't follow inverse square law because of	a. Its monochromaticity b. Its coherent c. Its non-divergence d. All of the above	
7	the common feature of laser and (x) rays photons are that they	a. Are coherent b. Are monochromatic c. Have the same speed d. Have the same energy	
8	the velocity of laser beam is the velocity of ordinary light	a. Higher than b. Equal to c. Lower than	
9	coherent photons in the optical rays means	a. Emitted by variable phase difference b. Moving in parallel beam c. Emitted by constant phase difference d. Doesn't obey inverse square law	
10	if the intensity of a laser ray 10cm away from its source is (I), so its intensity value away from 20cm is	a. I/4 b. I/2 c. I d. 2I	

11	from the main components of laser	a. Electrons b. Holes c. Active medium	
12	population inversion is state of the system, where in the upper energy state compared to lower energy state is populated	a. Less than b. More than c. Equal d. None of the above	
13	The lowest energy state is called	a. Ground states. b. Excited states. c. All the above.	
	The highest energy state is called	a. Ground states. b. Excited states. c. All the above.	

2-Mention the application of

No.	Question	Answer
1	active medium in laser	
2	Source of energy in laser	
3	Resonant cavity in laser	

3-Mention the scientific term

No.	Question	Answer
1	the acronym for Light Amplification by Stimulated Emission of Radiation	
2	Occurs when the atom relaxes from an excited state to a lower state after lifetime interval is over (nearly 10^{-8} s) emitting spontaneously the energy difference in the form of a photon without the effect of an external photon	
3	Occurs where an external photon of energy $E_2 - E_1$ passes by excited atom at E_2 to relax it to E_1 before lifetime is over	

4	the laser emits energy over a very narrow wavelength range, literally, one color.	
5	The diameter of the emitted light beam stays constant for long distance without much un scattering Thus, energy is transmitted without much loss.	
6	the emitted Photons are in phase and in the same direction	
7	Photons emanate coherently both in time and place, since they come out together at the same time sequence	
8	The number of atoms in the excited state exceeds the number of atoms in the lower	
9	the Photons are increased in number going back and forth in the active medium, due to multiple reflections between two enclosing mirrors	
10	The effective material required to produce laser	
11	it is the responsible for exciting the active medium	
12	it is the container and the activating catalyst for amplification	
13	It is an excited energy level characterized by its lifetime is relatively long nearly 10^{-3} s	

4-Solve the following:

No.	Question	Answer
1	A laser emits light with difference in two energy levels E_2 & E_1 is 3 eV then the wavelength of the radiation in nm is (to convert eV to J , multiply eV in the charge of the electron)	